Vignesh Kumar

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Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, Bash, Go, C#. **Databases**: SQL (PostgreSQL, Oracle), NoSQL (MongoDB, Cassandra, Redis), Graph DB (Neo4j) **Ops**: AWS (EC2, S3, RDS, Lambda, EKS), Azure, GCP, Docker, Kubernetes, Terraform, Jenkins, Git **Automation**: Selenium WebDriver, Playwright, WebDriverIO, PyTest, RestAssured, JMeter, TestNG, Cucumber, POM **AI** : Cursor, GitHub Copilot, Claude Code, Meta Llama 3, Prompt Engineering **Tools**: Splunk, ELK Stack (Elasticsearch, Logstash, Kibana), Grafana, CloudWatch, Postman, Swagger, Wireshark **Methodologies**: CI/CD Pipelines, Agile (Scrum, Kanban), API Testing, Performance Testing, Security Testing

Certifications & Awards

Certifications: AWS Certified Solutions Architect, Cisco Certified Network Associate & Professional (CCNA, CCNP), ISTQB, CompTIA PenTest+
Awards: Won *Top Contributor Award* for impactful code reviews across 2 consecutive quarters at Okta Inc.

Professional Experience

Freelance Consultant & AI Automation Contributor **Remote**
Upwork Inc. *April 2025 – Present*

- Prototyped web UI and REST API testing frameworks using Playwright, TypeScript, and reusable page objects.
- Generated API tests using Meta Llama 3, Python, and Neo4j, boosting coverage by 60% and cutting creation time by 50%.
- Utilized Cursor and Claude Code for test scaffolding, validating AI-generated code against enterprise reliability standards.
- Prototyped a TypeScript utility using DOM context to recommend optimal Playwright locators and test steps.

Finra (Contract) **Remote**
Senior Software Engineer in Test *Jan 2025 – Mar 2025*

- Built a Playwright framework with self-healing selectors to migrate from Selenium, improving test stability by 80%.
- Maintained end-to-end UI tests using the Page Object Model (POM) to eliminate flakiness from dynamic DOM elements.
- Integrated automated smoke tests into Jenkins, reducing per-PR execution times from 2 hours to 15 minutes.

Okta Inc. **Remote**
Senior Software Engineer in Test - Lead *Nov 2020 – Oct 2024*

- Executed testing strategies for large-scale frontend, API, and backend systems serving 3M+ clients and 19k+ organizations.
- Led automation for Multibrand and Risk Engine offerings, driving developer UX and increasing feature adoption by 27%.
- Built API regression tests for Multi-Branding workflows, covering certificate provisioning across isolated tenants.
- Combined Selenium tests with APIs to validate MFA and access blocking policies under configured user risk conditions.
- Engineered a PyTest workflow using AWS S3 and syslog parsing to validate Breached Credentials, logging 12 critical defects.
- Ran distributed JMeter performance tests on AWS to validate Threat Insight behavior and monitor latency at scale.
- Built Splunk monitoring dashboards for core functional metrics, reducing mean time to detection (MTTD) by 30%.
- Mentored cross-functional developers and test engineers to clear technical roadblocks and ensure timely project delivery.

RSA Security **Reston, VA / Remote**
Software Engineer II - Quality *Jul 2017 – Sep 2020*

- Implemented a UI automation test suite using WebDriver.io to capture defects early, reducing regression overhead by 40%.
- Migrated 200+ legacy tests to WebDriver.io, adding explicit waits for WebSocket UI components to eliminate test flake.
- Engineered Python/Pytest backend automation to validate NetWitness OS builds, halving overall test suite execution time.
- Built a 15-minute Jenkins micro-CI pipeline for critical smoke tests, catching failures before main branch integration.
- Validated OS deployments across distributed servers using CLI, logs, and network configurations for rapid troubleshooting.
- Troubleshot packet and log ingestion across physical and VMware environments using Wireshark, curl, and Linux/Bash.

Forbes Media **New Jersey**
Summer Intern *Jun 2016 – Dec 2016*

- Contributed 300+ JavaScript classes (Protractor) to increase coverage for a dynamic AngularJS application.
- Developed a Python broken-link crawler to track web services and generate automated downtime reports.

Zoho Corporation **India**
Member of Technical Staff - QA *Dec 2014 – Jul 2015*

- Tested REST API endpoints using Postman and Newman for the Zoho Mail application, supporting 300M+ users.
- Provisioned Docker containers on CentOS servers to establish unified test env and enhance cross-team collaboration.
- Validated user stories, identified defects, and triaged critical issues across multiple application release versions.

Bally Technologies **India**
Associate QA Engineer *Jun 2014 – Dec 2014*

- Performed functional and regression testing on Linux gaming systems, capturing critical bugs and fast regulatory approvals.
- Executed automated Python scripts for backend verification and core system logic validation.

Amazon.com **India**
Device Associate *Dec 2012 – Jun 2014*

- Executed functional and cross-platform compatibility tests across Amazon Kindle e-Ink and Android devices.
- Wrote custom Bash scripts to measure device speed and performance, resulting in an 18% improvement in user readability.

Education

New Jersey Institute of Technology **Newark, NJ**
Master of Science in Computer Science *May 2017*

Bharath University **Chennai, India**
Bachelor of Science in Computer Science *May 2012*